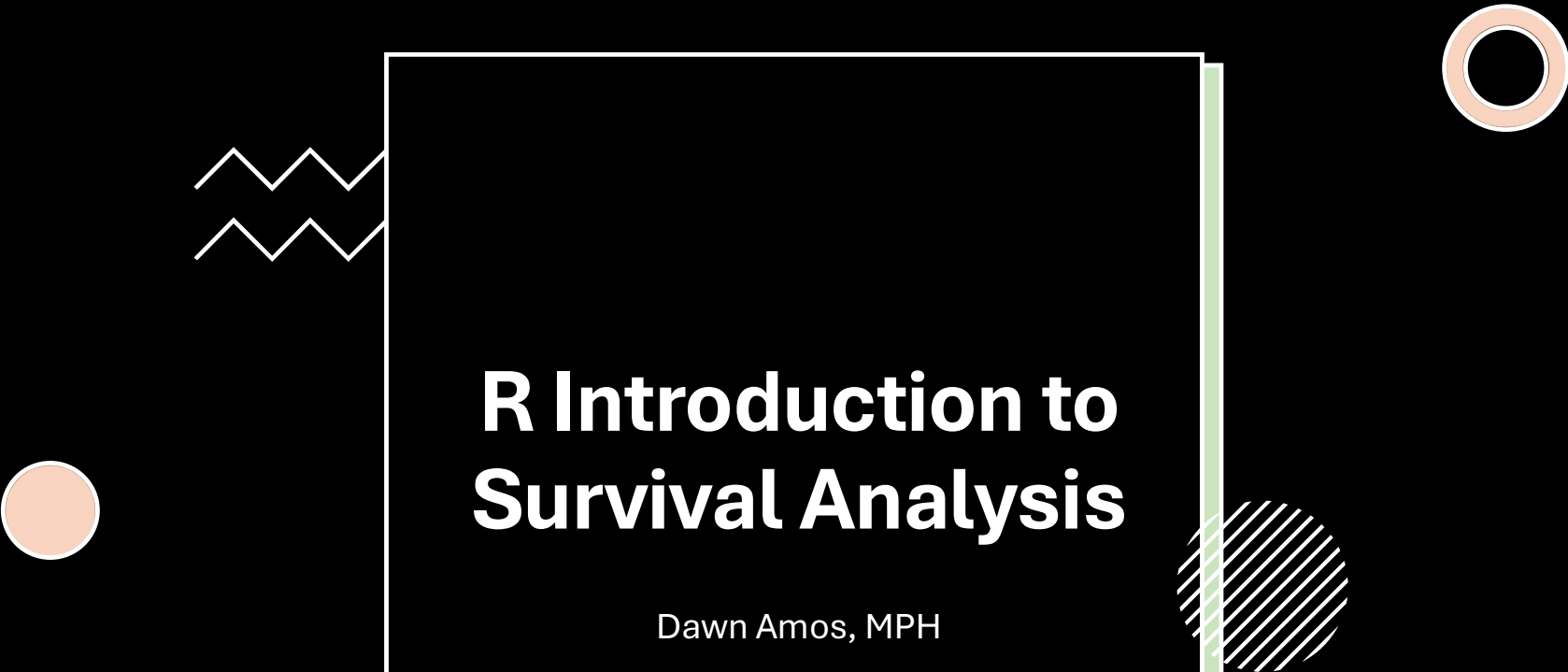


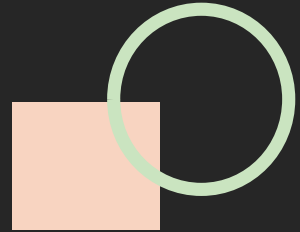


R Introduction to Survival Analysis

Dawn Amos, MPH

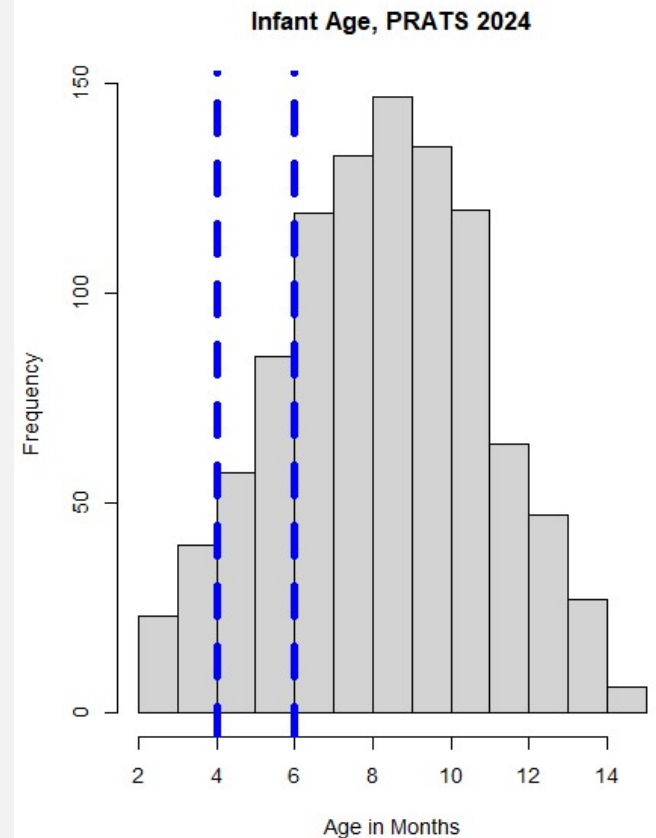
Idaho Department of Health and Welfare

The Problem: Infant Feeding

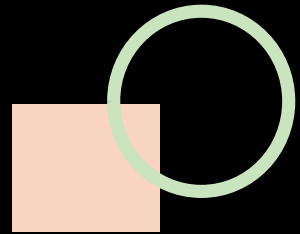


Study Question: What predicts when infants are introduced to solid foods?

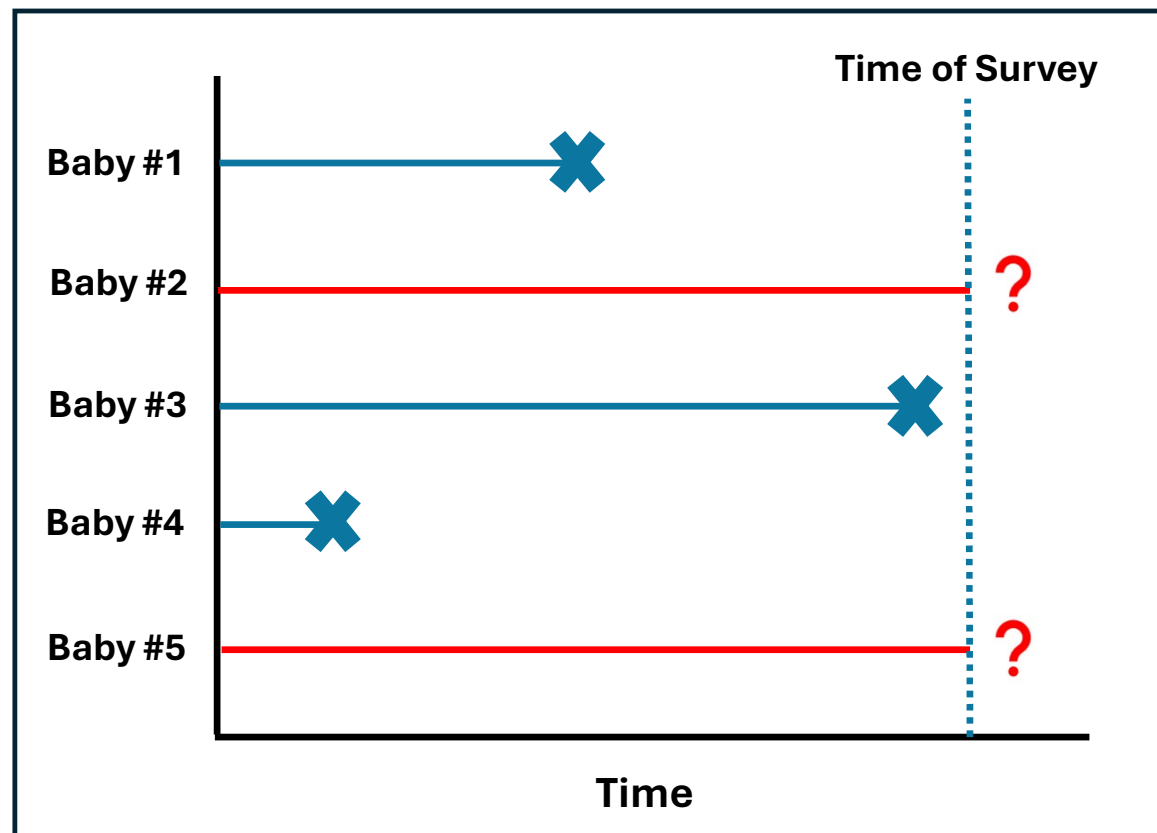
- Solids usually introduced at 4 – 6 months
- Sample
 - n = 877 (complete responses)
 - **Infant ages:** 2.6 to 15 months
 - **Never fed solids:** ~18% of sample (n = 154)



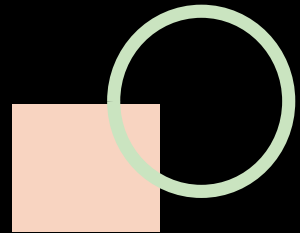
The Problem: Infant Feeding, cont.



- **Issue:** 18% of our sample contains censored data
- **Censored data:** exact outcome is unknown
 - Outcome occurs outside of the study window or outcome is loss to follow-up



The Problem: Infant Feeding, cont.

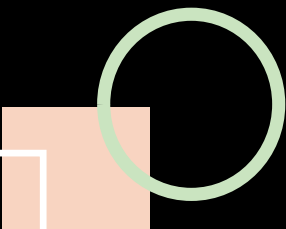


What do we use when we...

- Want to study what predicts when an event happens
- Have censored observations (unknown outcomes)

Answer:

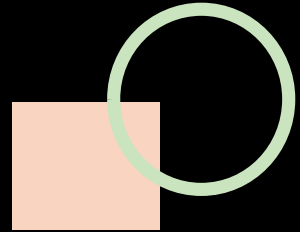
Survival (Time-to-Event) Analysis



Survival Analysis

1. What is a survival analysis?
2. Test and model assumptions
3. How to prep your data
4. Different types of survival analysis

Intro to Survival Analysis



Survival Analysis: analyzes the time to an event

Allows for the inclusion of censored data

- Unknown outcomes don't need to be removed
- Censored data not required

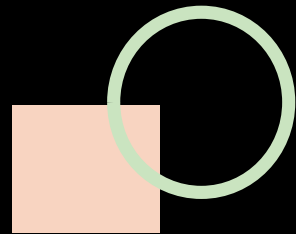
Most common survival analyses:

- **Kaplan-Meier method**
- **Log-Rank Test**
- **Cox proportional hazards model**

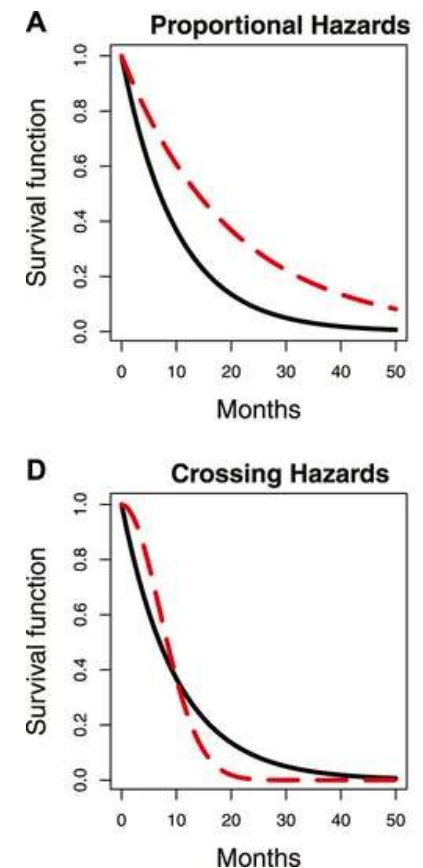


The Bear Grylls of
statistical analysis

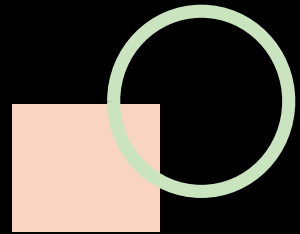
Assumptions



- 1. Independent observations** → survival of one observation is independent from the survival of other observations
- 2. Independent censoring** → reason for a subject being censored is unrelated to the event
- 3. Proportional hazards** → hazard ratio between two groups are constant over time



Prepping Your Data



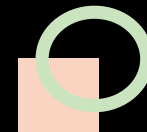
Using the R **survival** package

Two columns for each respondent:

- **Status** → binary variable (0-1) that indicates whether the event has happened to a respondent
- **Time**
 - If 0 ... time at study end
 - If 1 ... time of event



Prepping Your Data, cont.



Example:

Respondent	Fed solids	Infant age at study end (months)	Solid intro (months)	Status	Time
101	Yes	9	6	1	6
102	No	2.5	-	0	2.5
103	Yes	7	4.5	1	4.5
104	No	5	-	0	5

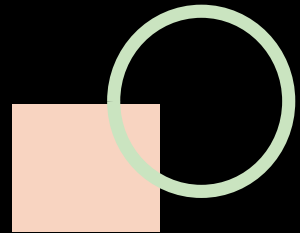
Green highlight: censored observation

Yellow highlight: complete observation

Practice Question:

Respondent	Fed solids	Infant age at study end (months)	Solid intro (months)	Status	Time
105	No	4	-		
106	No	3.5	-		
107	Yes	8	6.5		
108	Yes	10	5		

Prepping Your Data, cont.



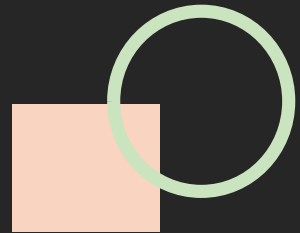
Practice Answer:

Respondent	Fed solids	Infant age at study end (months)	Solid intro (months)	Status	Time
105	No	4	-	0	4
106	No	3.5	-	0	3.5
107	Yes	8	6.5	1	6.5
108	Yes	10	5	1	5

Green highlight: censored observation

Yellow highlight: complete observation

Prepping Your Data, cont.



Input

```
library(dplyr)

# creating our variables for survival analysis
solids = solids %>%
  mutate(
    status = introduced_solids,
    time = case_when(
      status == 0 ~ infant_age_months,
      status == 1 ~ age_solids_introduced_months))

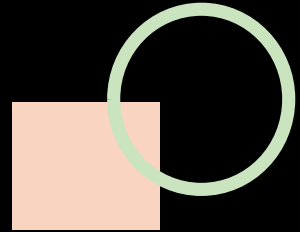
# checking our work
solids %>%
  select(infant_age_months,
         age_solids_introduced_months,
         time,
         status) %>%
  head()
```

Output

```
Prep Variables, output

  infant_age_months age_solids_introduced_months time status
1             7.6                      5.1  5.1         1
2             6.9                      NA  6.9         0
3             5.6                      NA  5.6         0
4             4.5                      4.5  4.5         1
5             6.5                      5.9  5.9         1
6             6.1                      NA  6.1         0
```

Types: Kaplan-Meier

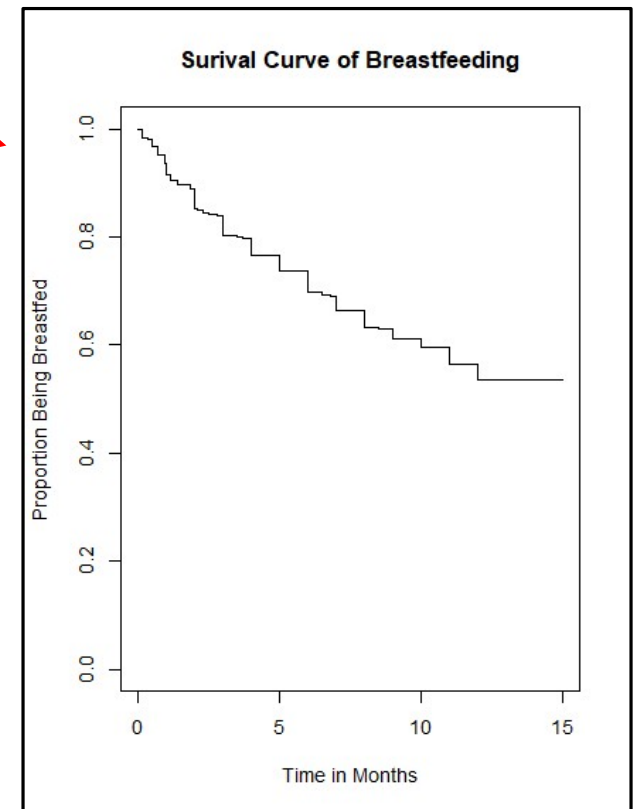


Kaplan-Meier: gives us a survival curve

- Plots time on the x-axis and proportion of surviving observations on the y-axis

Tells us...

- Proportion of surviving observations at a certain time point
- Probability of survival at a certain time point



Types: Kaplan-Meier, cont.



Kaplan Meier Curve

```
library(survival)

# Kaplan-Meier Curve
km_plot = survfit(Surv(time, status) ~ 1, data = solids)

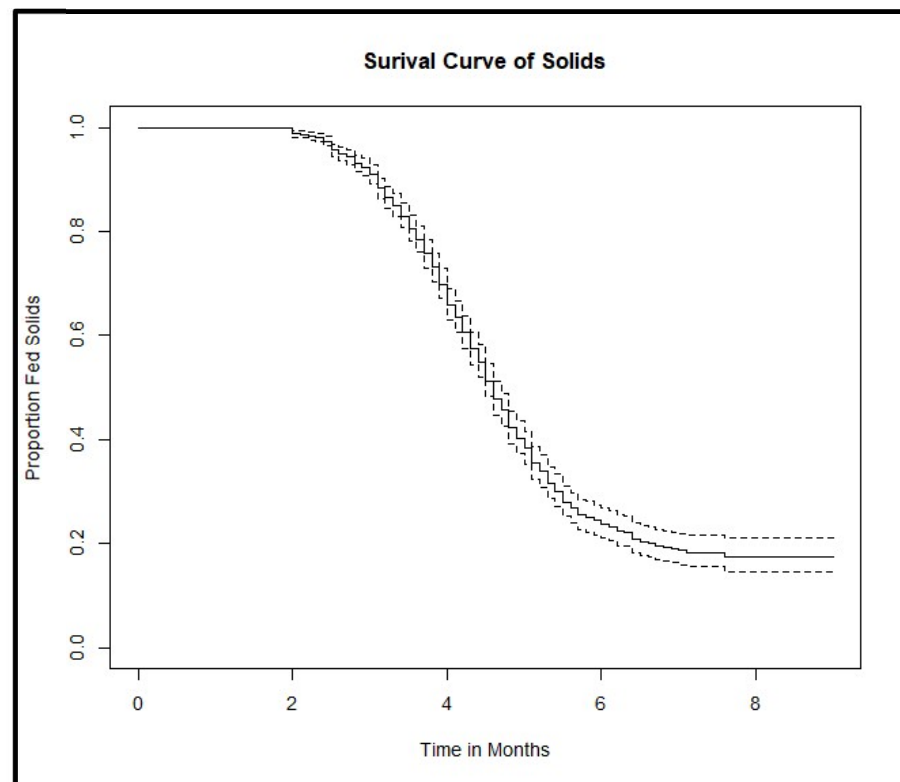
plot(km_plot,
     main = "Survival Curve of Solids",
     xlab = "Time in Months",
     ylab = "Proportion Fed Solids")

library(survey)
library(survminer)

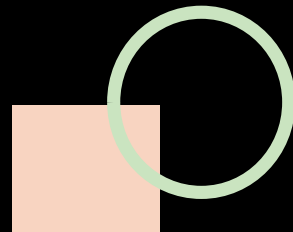
# creating a survey design
solids.design = svydesign(ids = ~1,
                        strata = ~ethnicity,
                        weights = ~survey_weight,
                        fpc = ~N,
                        data = solids)

# Kaplan-Meier Curve with a survey design (survey weights)
km_weighted_plot = svykm(Surv(time, status) ~ 1, design = solids.design)

plot(km_weighted_plot,
     main = "Survival Curve of Solids",
     xlab = "Time in Months",
     ylab = "Proportion Fed Solids")
```



Types: Log-Rank Test

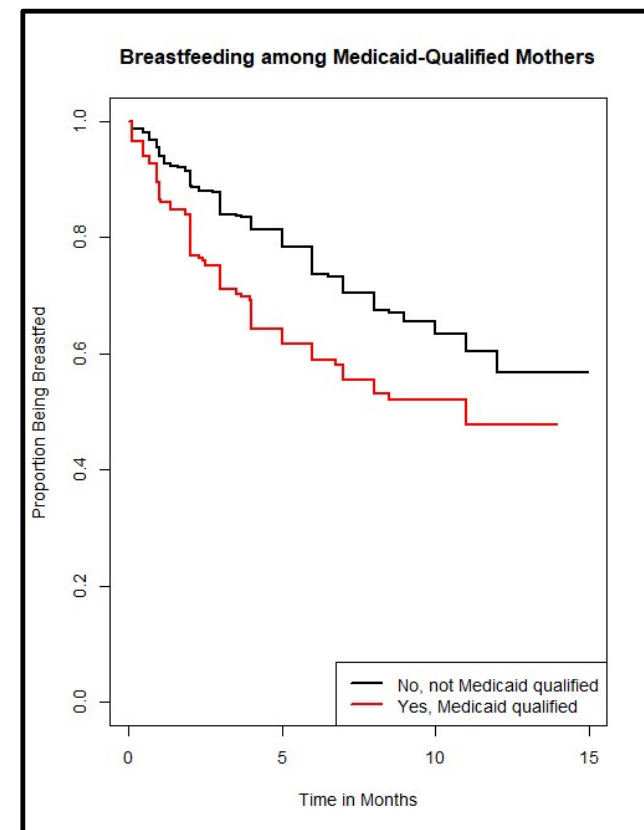


Log-Rank Test: compares the survival curve of two different groups

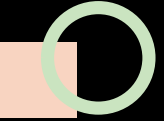
- Type of chi-squared test

Tells us...

- Are the survival curves of these two groups significantly different?



Types: Log-Rank Test, cont.



Input

```
Log-Rank

# Log-Rank Test
log_rank = survdiff(Surv(time, status) ~ insurance,
                    data = solids) %>%

  print()

# weighted Log-Rank Test
log_rank_weighted = svylogrank(Surv(time, status) ~ insurance,
                              design = solids.design) %>%

  print()
```

Output

```
Log-Rank, output

> # Log-Rank Test
> log_rank = survdiff(Surv(time, status) ~ insurance,
+                     data = solids) %>%
+   print()
Call:
survdiff(formula = Surv(time, status) ~ insurance, data = solids)


```

	N	Observed	Expected	(O-E)^2/E	(O-E)^2/V
insurance=Private	472	337	356.1	1.0208	2.1066
insurance=Medicaid	414	288	291.8	0.0494	0.0867
insurance=None	114	95	72.1	7.2454	8.3896

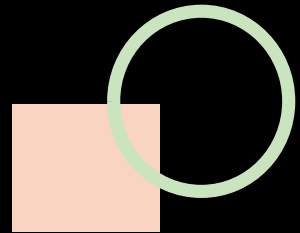
```

Chisq= 8.7 on 2 degrees of freedom, p= 0.01
>
> # weighted Log-Rank Test
> log_rank_weighted = svylogrank(Surv(time, status) ~ insurance,
+                               design = solids.design) %>%
+   print()
[[1]]
      score      se      z      p
[1,] -106.4852 261.4229 -0.4073292 0.683766188
[2,]  453.2192 155.8228  2.9085552 0.003631031

[[2]]
      Chisq      p
8.7599772 0.0125255

attr(,"class")
[1] "svylogrank"
```

Types: Log-Rank Test, cont.

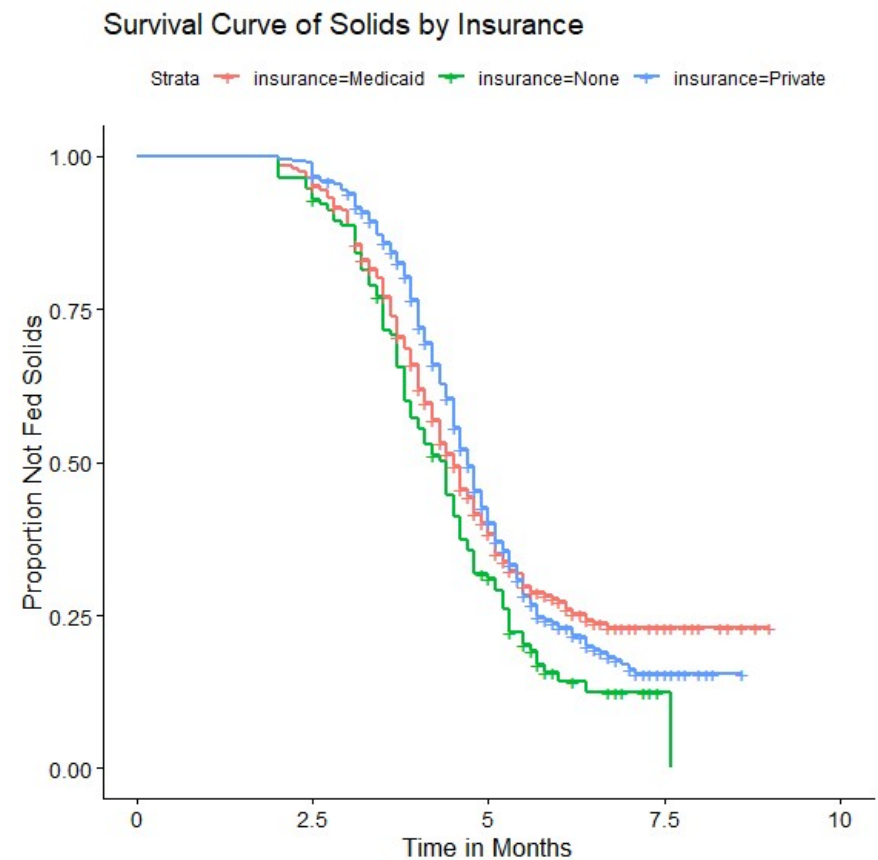


Un-Weighted Data

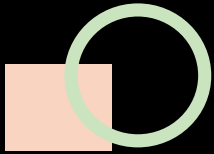
```
Plotting Log-Rank

# create fit first
insurance_fit = survfit(Surv(time, status) ~ insurance,
                        data = solids)

# using ggsurvplot()
ggsurvplot(insurance_fit,
            title = "Survival Curve of Solids by Insurance",
            xlab = "Time in Months",
            ylab = "Proportion Not Fed Solids")
```



Types: Log-Rank Test, cont.

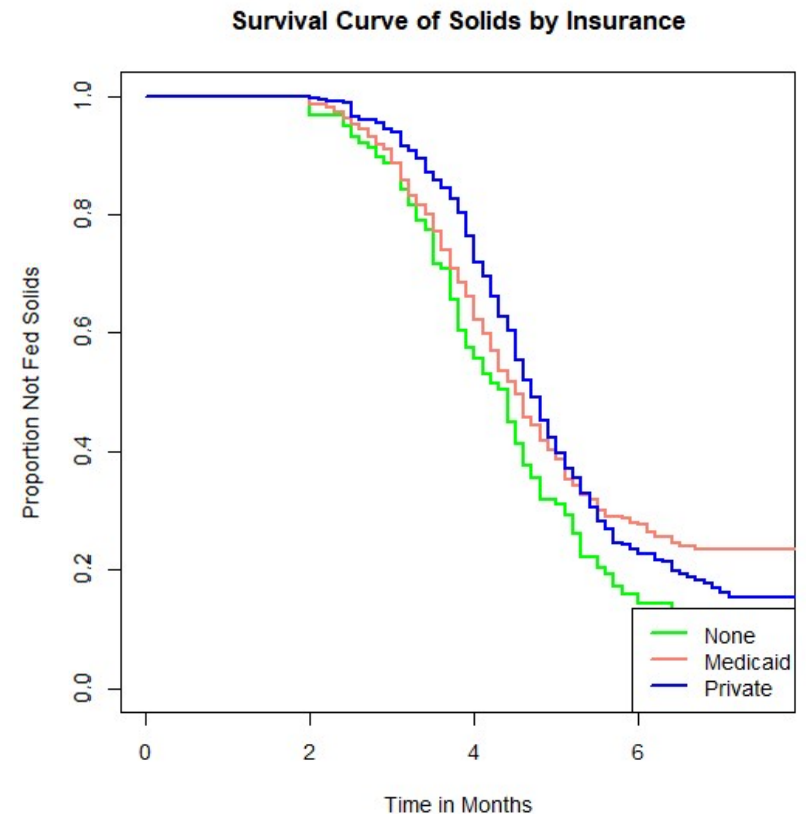


Weighted Data

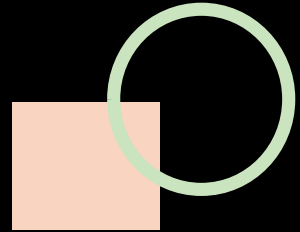
```
Plotting Log-Rank

# created weighted fit first
insurance_fit_weighted = svykm(Surv(time, status) ~ insurance,
                              design = solids.design)

# using plot()
plot(insurance_fit_weighted[["None"]],
     main = "Survival Curve of Solids by Insurance",
     xlab = "Time in Months",
     ylab = "Proportion Not Fed Solids",
     col = "green", lwd = 2)
lines(insurance_fit_weighted[["Medicaid"]], col = "salmon", lwd = 2)
lines(insurance_fit_weighted[["Private"]], col = "blue", lwd = 2)
# Add a legend
legend("bottomright",
      legend = c("None", "Medicaid", "Private"),
      col = c("green", "salmon", "blue"),
      lty = 1,
      lwd = 2)
```



Types: Cox Proportional Hazards Model



Cox Proportional Hazards Model: assesses how different predictors impact the time it takes for an event to occur

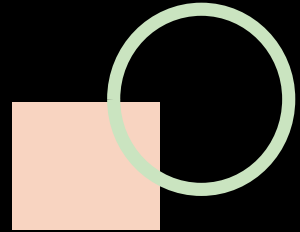
- Gives us hazard ratios

Tells us...

- Does this predictor speed-up or slow-down the rate of event occurrence?



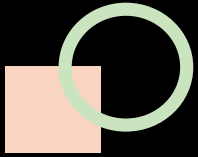
Types: Cox Proportional Hazards Model



A ***hazard ratio*** is...

- A rate of occurrence
- Comparative → relative to the reference group
- Instantaneous → at given moment in the study period

Types: Cox Proportional Hazards Model



Imagine we have two buildings.
Each building contains with 1,000 lightbulbs.

Building A

Each hour, 2 lightbulbs burn out.



Rate: 2 per hour

Building B

Each hour, 4 lightbulbs burn out.



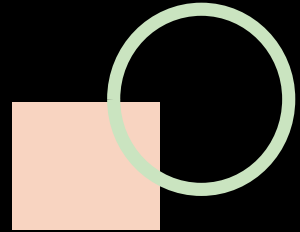
Rate: 4 per hour

Hazard Ratio

Building B's **rate of occurrence** is 2 times that of
Building A's rate of occurrence in **any given moment**.

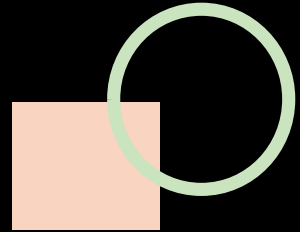
- **Rate of occurrence**
- **Relative to reference group**
- **At any given moment**

Types: Cox Proportional Hazards Model



Hazard Ratio (HR) Value	Interpretation	Example
HR = 1.0	No difference	"This group's rate of occurrence at any given moment is the same as the reference group."
HR < 1.0	Lower hazard → slower event occurrence	"This group's rate of occurrence is 0.5 times that of the reference group at any given moment."
HR > 1.0	Higher hazard → faster event occurrence	"This group's rate of occurrence is 1.5 times that of the reference group at any given moment."

Types: Cox Proportional Hazards Model



Un-Weighted Data

Input

```
Un-Weighted Cox Proportional Hazards Model

cox_model = coxph(Surv(time, status) ~ insurance +
ethnicity + maternal_age,
                  data = solids) %>%
print()
```

Input

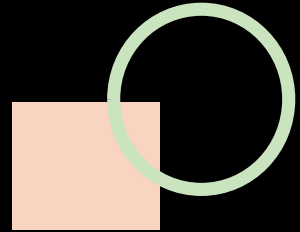
```
Un-Weighted Cox Proportional Hazards Model, output

> cox_model = coxph(Surv(time, status) ~ insurance + ethnicity +
maternal_age,
+                  data = solids) %>%
+   print()
Call:
coxph(formula = Surv(time, status) ~ insurance + ethnicity +
      maternal_age, data = solids)

              coef exp(coef) se(coef)      z      p
insuranceMedicaid  0.042084  1.042982  0.080584  0.522 0.60151
insuranceNone       0.345588  1.412820  0.116291  2.972 0.00296
ethnicityHispanic   0.058476  1.060219  0.089651  0.652 0.51423
maternal_age       -0.023096  0.977169  0.007532 -3.066 0.00217

Likelihood ratio test=17.64 on 4 df, p=0.001453
n= 1000, number of events= 720
```

Types: Cox Proportional Hazards Model



Weighted Data

Input

```
Weighted Cox Proportional Hazards Model

cox_model_weighted = svycoxph(Surv(time, status) ~ insurance +
  ethnicity + maternal_age,
  design = solids.design) %>%
  print()
```

Input

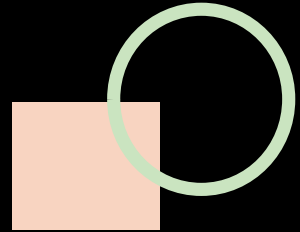
```
Weighted Cox Proportional Hazards Model, output

> cox_model_weighted = svycoxph(Surv(time, status) ~ insurance +
  ethnicity + maternal_age,
  + design = solids.design) %>%
  + print()
Call:
svycoxph(formula = Surv(time, status) ~ insurance + ethnicity +
  maternal_age, design = solids.design)

      coef exp(coef) se(coef) robust se      z      p
insuranceMedicaid  0.031373  1.031870  0.080608  0.080293  0.391 0.69600
insuranceNone       0.338935  1.403452  0.116407  0.114642  2.956 0.00311
ethnicityHispanic   0.057604  1.059295  0.094099  0.088061  0.654 0.51302
maternal_age       -0.023337  0.976933  0.007531  0.007848 -2.974 0.00294

Likelihood ratio test= on 4 df, p=
n= 1000, number of events= 720
```

Types: Cox Proportional Hazards Model



```
Weighted Cox Proportional Hazards Model, output

> cox_model_weighted = svycoxph(Surv(time, status) ~ insurance +
+ ethnicity + maternal_age,
+                               design = solids.design) %>%
+   print()
Call:
svycoxph(formula = Surv(time, status) ~ insurance + ethnicity +
  maternal_age, design = solids.design)

      coef exp(coef) se(coef) robust se      z      p
insuranceMedicaid  0.031373  1.031870  0.080608  0.080293  0.391 0.69600
insuranceNone       0.338935  1.403452  0.116407  0.114642  2.956 0.00311
ethnicityHispanic   0.057604  1.059295  0.094099  0.088061  0.654 0.51302
maternal_age        -0.023337  0.976933  0.007531  0.007848 -2.974 0.00294

Likelihood ratio test=  on 4 df, p=
n= 1000, number of events= 720
```

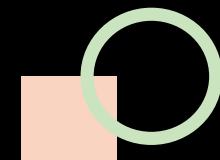
Model: predicting time of solid food introduction by insurance type, ethnicity, and maternal age

Findings:

- At any given moment, mothers with no insurance introduced solids at a rate that is 1.4 times that of mothers with private insurance ($p < 0.01$).
 - Faster time-to-event
- For every year increase in age, the rate of introducing solids is 0.98 times that of mothers one year younger ($p < 0.01$).
 - Slower time-to-event

Number of non-censored observations

Recap of Content



- **Survival analysis** analyzes the time to an event.
- Survival analyses can handle censored (incomplete) observations.
- To use R's `survival` package, you need to create a status and time variable.
- The three most common forms of survival analysis are...
 - **Kaplan-Meier Method:** gives you a survival curve
 - **Log-Rank Test:** compares the survival curve of different groups to determine statistical significance
 - **Cox Proportional Hazards Model:** allows you to see how certain predictors speed up or slow down the rate of occurrence for an event
 - **Hazard ratios** are relative rates of occurrence for a any single unit of time

Thanks for listening!